Sora Kim

HCI Researcher · Human-AI Interaction and Accountability in High-Stakes Systems

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RESEARCH FOCUS

I study how accountability is structured in regulated work and how system design shapes which causal explanations remain available before responsibility settles. My research combines HCI experimentation with STS analysis, using pharmaceutical GMP environments as a primary empirical site. Intervention work examines human-AI interaction as a reasoning intervention: resisting premature closure and preserving the comparability of alternative explanations before attribution stabilizes.

RESEARCH

HCI

Keeping Comparison Open: Interface Scaffolding Against Premature Accountability Convergence in Regulated Investigations

Preprint, 2026 · CSCW 2026 poster under review

N=20 + N=20. Study 1: 18/20 participants broadened causal reasoning (coded causal breadth expansion). Study 2: 15/20 reconsidered initial attribution after scaffolded comparison (attribution shift protocol).

How Regulated Systems Produce Responsibility: A Four-Layer Framework for Interaction Design

Preprint, 2026

Introduces Regulatory UX — a framework synthesizing findings across the HCI project set into a unified account of how system design shapes accountability conditions under human-AI interaction.

From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work

Preprint, 2026

N=20, 500 synthetic GMP SOPs. Reduced retrieval time ∼6 min → ∼1 min (task-timed); surfaced invisible onboarding labor upstream of formal quality records.

When Documents Don't Connect: Traceability Burden and Loopback Behavior in GMP Document Navigation

Preprint, 2026

Graph-based traceability visibility and behavioral loopback tracking. 30–60% reduction in path reconstruction time (session-logged).

Friction Before Failure: Making Review Burden Visible in Regulated Document Work

Preprint, 2026

Six-tab dashboard and TF-IDF pattern detection over 100 synthetic deviation-CAPA records. Surfaces review burden invisible to existing quality metrics.

STS — Theoretical Foundations

Anticipated Accountability Convergence: Structuring Responsibility Before Action

Preprint, 2026

Structuring Judgment Under Constraint: Cognitive Friction in Procedural Texts

Preprint, 2026

Responsibility Convergence in Organizational Investigations

Preprint, 2026

Premature Accountability Convergence in Regulated Organizations

Preprint, 2026 · 4S Toronto 2026 under review

From GMP to High-Stakes Systems: Accountability Convergence as a Cross-Domain Phenomenon in Regulated Investigations

Manuscript in preparation, 2026

PROFESSIONAL EXPERIENCE

Quality Control / Quality Systems Project Lead — Sep 2022 – Present

Prestige Biologics, South Korea

- Led quality system operations across a 30-person QC/QA team, managing 20+ deviation and CAPA investigations end-to-end in GMP-regulated pharmaceutical manufacturing.
- Supported 10+ regulatory inspections and global audits; developed working expertise in how documentation architecture shapes traceability, coordination, and responsibility attribution.
- Revised and maintained 50+ SOPs across validation, deviation, CAPA, and RCA workflows, directly observing how procedural document design produces accountability patterns.
- Operated LIMS, ELN, and Veeva Vault in daily GMP practice; this environment served as the primary empirical site for the current HCI research program.

METHODS & SKILLS

Methods	Experimental design (within-subject, task-based); think-aloud protocol; behavioral logging; causal reasoning analysis; annotation protocol design; qualitative coding (open, axial); document and corpus analysis; NLP / text processing (TF-IDF, sentence-level annotation)
Tools	Python · R · Streamlit · LaTeX · HTML/CSS · GitHub
Systems	LIMS · ELN · Veeva Vault
Languages	Korean (native) · English (fluent) · Japanese (fluent)

Last Updated: March 2026